

DIGITAL SAT MATH — COMPLETE SOLUTIONS

DOCUMENT 2 | FULL EXPLANATIONS & SOLUTIONS

Topics: Function Notation & Evaluation | Creating Linear Models from Word Problems

50 Questions | Step-by-Step Solutions | Trap Alerts | Key Insights | Fast Methods

SECTION A — FUNCTION NOTATION & EVALUATION (Q1–Q25)

Q1. Correct Answer: **B** [EASY]

Context: $h(x) = 7x + 3$; find $h(5)$

STEP-BY-STEP SOLUTION:

1. Substitute $x = 5$ into $h(x) = 7x + 3$.
2. $h(5) = 7(5) + 3 = 35 + 3 = 38$.

WHY EACH OPTION IS WRONG:

- A)** 35 — Stopped after multiplying: forgot to add the constant +3.
- B)** 38 — CORRECT: $7(5) + 3 = 35 + 3 = 38$. ✓
- C)** 43 — Added 8 instead of 3, or multiplied incorrectly.
- D)** 40 — Treated the coefficient as 8 instead of 7.

■ **TRAP ALERT:** Stopping at 35 after the multiplication and forgetting to add +3.

■ **KEY INSIGHT:** $f(a)$: substitute the number everywhere x appears, then simplify step by step.

■ **FAST METHOD:** $7 \times 5 = 35$, $+ 3 = 38$. Two steps.

Q2. Correct Answer: **B** [EASY]

Context: $f(x) = 4x^2 - 2x + 1$; find $f(0)$

STEP-BY-STEP SOLUTION:

1. Substitute $x = 0$.
2. $f(0) = 4(0)^2 - 2(0) + 1 = 0 - 0 + 1 = 1$.

WHY EACH OPTION IS WRONG:

- A)** 0 — Ignored the constant term +1.
- B)** 1 — CORRECT: every term with x vanishes; only the constant survives. ✓
- C)** -1 — Changed the sign of the constant.
- D)** 3 — Evaluated by setting $x = 1$ instead of $x = 0$.

■ **TRAP ALERT:** When $x = 0$, all terms containing x disappear — only the constant remains.

■ **KEY INSIGHT:** $f(0)$ always equals the constant term of any polynomial (the y -intercept).

■ **FAST METHOD:** Plug in 0: every x -term vanishes. Only $+1$ remains.

Q3. Correct Answer: B [EASY]

Context: $g(t) = 12 - 3t$; find $g(4)$

STEP-BY-STEP SOLUTION:

1. Substitute $t = 4$: $g(4) = 12 - 3(4)$.
2. $= 12 - 12 = 0$.

WHY EACH OPTION IS WRONG:

- A) 24 — Added instead of subtracted: $12 + 12$.
- B) 0 — CORRECT: $12 - 12 = 0$. ✓
- C) 6 — Subtracted only 3 once instead of computing $3 \times 4 = 12$.
- D) -2 — Arithmetic error: subtracted 14 instead of 12.

■ **TRAP ALERT:** Subtracting 3 instead of the product $3 \times 4 = 12$.

■ **KEY INSIGHT:** Order of operations: multiply 3×4 first, then subtract from 12.

■ **FAST METHOD:** $3 \times 4 = 12$. $12 - 12 = 0$.

Q4. Correct Answer: B [EASY]

Context: $p(x) = 5x - 9$; solve $p(x) = 16$

STEP-BY-STEP SOLUTION:

1. Set $5x - 9 = 16$.
2. Add 9 to both sides: $5x = 25$.
3. Divide by 5: $x = 5$.

WHY EACH OPTION IS WRONG:

- A) 3 — Used wrong operations; check: $5(3) - 9 = 6 \neq 16$.
- B) 5 — CORRECT: $5(5) - 9 = 25 - 9 = 16$. ✓
- C) 7 — Added 9 incorrectly; check: $5(7) - 9 = 26 \neq 16$.
- D) 4 — Check: $5(4) - 9 = 11 \neq 16$.

■ **TRAP ALERT:** Forgetting to add 9 to both sides before dividing.

■ **KEY INSIGHT:** To find x given $f(x) = \text{value}$, set the expression equal to the value and solve like a standard equation.

■ **FAST METHOD:** $5x = 16 + 9 = 25 \rightarrow x = 5$. Verify: $5(5) - 9 = 16$. ✓

Q5. Correct Answer: C [MEDIUM]

Context: $f(x) = 3x^2 + kx - 4$; $f(2) = 14$; find k

STEP-BY-STEP SOLUTION:

1. Substitute $x = 2$: $3(2)^2 + k(2) - 4 = 14$.
2. Simplify: $12 + 2k - 4 = 14$.
3. Combine constants: $8 + 2k = 14$.
4. Solve: $2k = 6 \rightarrow k = 3$.

WHY EACH OPTION IS WRONG:

- A) 0 — Forgot the constant -4 when simplifying.
- B) 2 — Arithmetic error: set $2k = 4$ instead of $2k = 6$.
- C) 3 — CORRECT: $3(4) + 3(2) - 4 = 12 + 6 - 4 = 14$. ✓
- D) 5 — Added 4 to both sides instead of accounting for -4 correctly.

■ **TRAP ALERT:** Moving -4 to the right side: it becomes $+4$, giving $2k = 6$, not $2k = 10$.

■ **KEY INSIGHT:** Substitute the given x -value, simplify all constants on the left, then isolate the unknown.

■ **FAST METHOD:** $f(2) = 12 + 2k - 4 = 8 + 2k = 14 \rightarrow 2k = 6 \rightarrow k = 3$.

Q6. Correct Answer: A [MEDIUM]

Context: $g(x)=2x-5$; $h(x)=x^2+1$; find $g(h(3))$

STEP-BY-STEP SOLUTION:

1. Step 1 — evaluate the inner function first: $h(3) = (3)^2 + 1 = 10$.
2. Step 2 — plug that result into the outer function: $g(10) = 2(10) - 5 = 15$.

WHY EACH OPTION IS WRONG:

- A) 15 — CORRECT: $h(3) = 10$, then $g(10) = 15$. ✓
- B) 5 — Computed $g(3)$ instead of $g(h(3))$: $2(3) - 5 = 1$ (different error path).
- C) 17 — Applied the functions in the wrong order (g first, then h).
- D) 11 — Correctly found $h(3) = 10$, but then computed $10 + 1 = 11$ instead of $g(10)$.

■ **TRAP ALERT:** Applying functions in the wrong order. $g(h(3))$ means evaluate h FIRST, then g .

■ **KEY INSIGHT:** Composition rule: work inside-out. $g(h(3)) \rightarrow$ compute $h(3) \rightarrow$ plug result into g .

■ **FAST METHOD:** $h(3) = 10$. $g(10) = 20 - 5 = 15$.

Q7. Correct Answer: A [MEDIUM]

Context: $f(x) = (x+3)^2 - 7$; find $f(-1)$

STEP-BY-STEP SOLUTION:

1. Substitute $x = -1$: $f(-1) = (-1 + 3)^2 - 7$.
2. Simplify inside the parentheses first: $(2)^2 - 7$.

3. $= 4 - 7 = -3$.

WHY EACH OPTION IS WRONG:

- A) -3 — CORRECT: $(2)^2 - 7 = 4 - 7 = -3$. ✓
- B) -7 — Computed $(-1)^2 + 3 - 7$ incorrectly, treating the grouping wrong.
- C) 0 — Used $(-1)^2 = 1$, then $1 + 3 - 7 = -3$; sign error in next step.
- D) 4 — Found $(2)^2 = 4$ correctly but forgot to subtract 7.

■ **TRAP ALERT:** Substituting -1 into x^2 before adding 3. The parentheses group $(x + 3)$ first.

■ **KEY INSIGHT:** Compute the entire expression inside the parentheses before squaring.

■ **FAST METHOD:** $(-1 + 3) = 2$. Then $2^2 - 7 = 4 - 7 = -3$.

Q8. Correct Answer: D [MEDIUM]

Context: $f(x)=8-2x$; $g(x)=3x+1$; solve $f(x)=g(x)$

STEP-BY-STEP SOLUTION:

1. Set $8 - 2x = 3x + 1$.
2. Move x terms to one side: $8 - 1 = 3x + 2x$.
3. $7 = 5x$.
4. $x = 7/5 = 1.4$.

WHY EACH OPTION IS WRONG:

- A) $x = 1$ — Verify: $f(1) = 6$, $g(1) = 4$. Not equal. ✗
- B) $x = 2$ — Verify: $f(2) = 4$, $g(2) = 7$. Not equal. ✗
- C) $x = -1$ — Verify: $f(-1) = 10$, $g(-1) = -2$. Not equal. ✗
- D) $x = 7/5$ — CORRECT: $f(7/5) = 8 - 14/5 = 26/5$; $g(7/5) = 21/5 + 5/5 = 26/5$. ✓

■ **TRAP ALERT:** Moving x -terms to the wrong side, or avoiding the fractional answer. The DSAT regularly produces non-integer solutions.

■ **KEY INSIGHT:** Solve $f(x) = g(x)$ algebraically, then verify by substituting back into both expressions.

■ **FAST METHOD:** $8 - 2x = 3x + 1 \rightarrow 7 = 5x \rightarrow x = 7/5$.

Q9. Correct Answer: A [MEDIUM]

Context: $h(n) = n^2 - 4n + 4$; which is equivalent?

STEP-BY-STEP SOLUTION:

1. Recognize the perfect square trinomial pattern: $a^2 - 2ab + b^2 = (a - b)^2$.
2. Here $a = n$, $b = 2$: $n^2 - 2(n)(2) + 4 = (n - 2)^2$.

WHY EACH OPTION IS WRONG:

- A) $(n - 2)^2$ — CORRECT: expands to $n^2 - 4n + 4$. ✓

B) $(n + 2)^2$ — Expands to $n^2 + 4n + 4$ (wrong sign on middle term).

C) $(n - 4)^2$ — Expands to $n^2 - 8n + 16$ (wrong coefficients).

D) $n(n - 4)$ — Expands to $n^2 - 4n$ (missing the +4 constant).

■ **TRAP ALERT:** Confusing $(n - 2)^2$ with $(n + 2)^2$. Always check the sign of the middle term.

■ **KEY INSIGHT:** Memorize: $(a - b)^2 = a^2 - 2ab + b^2$. The middle term coefficient is $-2ab$.

■ **FAST METHOD:** What number squared = 4? $\rightarrow 2$. What gives $-4n$? $\rightarrow -2(n)(2)$. So $(n - 2)^2$.

Q10. Correct Answer: **B** [MEDIUM]

Context: $f(x)=ax+b$; $f(0)=2$; $f(3)=17$; find a

STEP-BY-STEP SOLUTION:

1. Use $f(0) = 2$: $a(0) + b = b = 2$, so $b = 2$.
2. Use $f(3) = 17$: $a(3) + 2 = 17 \rightarrow 3a = 15 \rightarrow a = 5$.

WHY EACH OPTION IS WRONG:

A) 3 — Divided 17 by 3 without subtracting b first.

B) 5 — CORRECT: $3(5) + 2 = 17$. ✓

C) 7 — Added b instead of subtracting: $3a = 19$.

D) 15 — Forgot to divide by 3 after isolating $3a = 15$.

■ **TRAP ALERT:** Forgetting to find b first using $f(0)$. Without $b = 2$, you cannot isolate a .

■ **KEY INSIGHT:** Given two conditions for a linear function, use one to find b (intercept), then the other to find a (slope).

■ **FAST METHOD:** $f(0) = b = 2$. Then $3a + 2 = 17 \rightarrow a = 5$.

Q11. Correct Answer: **B** [HARD]

Context: $f(x)=2x+3$; $g(x)=f(x+4)-f(x)$; find $g(7)$

STEP-BY-STEP SOLUTION:

1. Find $f(x + 4)$: $f(x + 4) = 2(x + 4) + 3 = 2x + 8 + 3 = 2x + 11$.
2. Compute the difference: $g(x) = (2x + 11) - (2x + 3) = 8$.
3. $g(x) = 8$ for ALL values of x , so $g(7) = 8$.

WHY EACH OPTION IS WRONG:

A) 4 — Only evaluated $f(4)$, not the full difference $f(x+4) - f(x)$.

B) 8 — CORRECT: g is a constant function equal to 8. ✓

C) 11 — Evaluated $f(11)$ alone without subtracting $f(7)$.

D) 3 — Took only the constant term from the formula.

■ **TRAP ALERT:** For any linear function $f(x) = mx + b$, the difference $f(x + c) - f(x) = mc$ (a constant). Plugging in $x = 7$ is unnecessary.

■ **KEY INSIGHT:** For a linear function, the difference over any fixed interval = slope $\times$ interval length.

■ **FAST METHOD:** Slope is 2, interval is 4: $2 \times 4 = 8$. No need to plug in $x = 7$.

Q12. Correct Answer: **A** [HARD]

Context: $f(x) = x^2 - 3$; $g(x) = 2f(x) - f(2x)$; find $g(3)$

STEP-BY-STEP SOLUTION:

1. Compute $f(3) = (3)^2 - 3 = 9 - 3 = 6$.
2. Compute $f(2 \cdot 3) = f(6) = (6)^2 - 3 = 36 - 3 = 33$.
3. $g(3) = 2f(3) - f(6) = 2(6) - 33 = 12 - 33 = -21$.

WHY EACH OPTION IS WRONG:

- A)** -21 — CORRECT: $2(6) - 33 = 12 - 33 = -21$. ✓
- B)** -6 — Used $f(6) = 9$ (forgot to square 6 correctly) or wrong arithmetic.
- C)** 6 — Computed only $f(3) = 6$ without applying the full expression for g .
- D)** 18 — Reversed the subtraction: $33 - 12 = 21$, then sign error.

■ **TRAP ALERT:** $f(2x) \neq 2f(x)$. Substituting $2x$ into f gives $f(2x) = (2x)^2 - 3 = 4x^2 - 3$, which is very different from $2(x^2 - 3)$.

■ **KEY INSIGHT:** Always substitute the entire argument into the function definition — never just multiply the output.

■ **FAST METHOD:** $f(3) = 6$. $f(6) = 33$. $g(3) = 12 - 33 = -21$.

Q13. Correct Answer: **C** [HARD]

Context: $f(2x) = 4f(x)$; $f(1) = 5$; find $f(4)$

STEP-BY-STEP SOLUTION:

1. Apply the rule with $x = 1$: $f(2 \cdot 1) = f(2) = 4 \cdot f(1) = 4(5) = 20$.
2. Apply the rule with $x = 2$: $f(2 \cdot 2) = f(4) = 4 \cdot f(2) = 4(20) = 80$.

WHY EACH OPTION IS WRONG:

- A)** 20 — This is $f(2)$, not $f(4)$. Stopped one step too early.
- B)** 40 — Used $2 \times f(2)$ instead of $4 \times f(2)$.
- C)** 80 — CORRECT: $f(4) = 4 \cdot f(2) = 4(20) = 80$. ✓
- D)** 160 — Applied the rule one extra time: that would be $f(8)$, not $f(4)$.

■ **TRAP ALERT:** Applying the rule too many or too few times. Build up step by step: $f(1) \rightarrow f(2) \rightarrow f(4)$.

■ **KEY INSIGHT:** Each time x doubles, the output multiplies by 4. Build up from the known value.

■ **FAST METHOD:** $f(1)=5$, $f(2)=20$, $f(4)=80$.

Q14. Correct Answer: B [HARD]

Context: $p(x)=(x-a)(x-b)$; $p(0)=12$; $p(1)=0$

STEP-BY-STEP SOLUTION:

1. $p(0) = (0-a)(0-b) = ab = 12$.
2. $p(1) = (1-a)(1-b) = 0$, so $a = 1$ or $b = 1$.
3. Test option B: $a = 1$, $b = 12 \rightarrow ab = 12$ ✓ and $p(1) = (1-1)(1-12) = 0 \cdot (-11) = 0$. ✓

WHY EACH OPTION IS WRONG:

- A)** $a=3$, $b=4$ — Product = 12 ✓ but $p(1) = (-2)(-3) = 6 \neq 0$. ✗
B) $a=1$, $b=12$ — CORRECT: $ab=12$ ✓ and $p(1)=0$. ✓
C) $a=2$, $b=6$ — Product = 12 ✓ but $p(1) = (-1)(-5) = 5 \neq 0$. ✗
D) $a=1$, $b=-12$ — Product = $-12 \neq 12$. ✗

■ **TRAP ALERT:** Only checking the product condition $p(0) = 12$ and ignoring $p(1) = 0$ (which requires $a = 1$ or $b = 1$).

■ **KEY INSIGHT:** $p(1) = 0$ means $x = 1$ is a root, so at least one of a or b must equal 1. Use this to eliminate choices immediately.

■ **FAST METHOD:** $p(1)=0 \rightarrow x=1$ is a root $\rightarrow$ one of a , b must equal 1. Only option B satisfies this.

Q15. Correct Answer: D [HARD]

Context: $f(x)=cx^2-5$; $f(3)=f(-3)+12$; find c

STEP-BY-STEP SOLUTION:

1. $f(3) = 9c - 5$ and $f(-3) = 9c - 5$ (since x^2 is even: $(-3)^2 = (3)^2$).
2. The equation becomes: $9c - 5 = (9c - 5) + 12$.
3. Simplifying: $0 = 12$ — a contradiction for any value of c .
4. Therefore no value of c can satisfy the equation.

WHY EACH OPTION IS WRONG:

- A)** 1 — Check: $f(3) = 4$, $f(-3) = 4$; $4 \neq 4 + 12$. ✗
B) 2 — Check: $f(3) = 13$, $f(-3) = 13$; $13 \neq 13 + 12$. ✗
C) 3 — Same issue: cx^2 is always even, so $f(3) = f(-3)$ for any c . ✗
D) No unique solution exists — CORRECT: the equation creates a contradiction for all c . ✓

■ **TRAP ALERT:** Trying to solve algebraically without recognizing that cx^2 is an even function, making $f(3) = f(-3)$ always true — so the condition $f(3) = f(-3) + 12$ can never hold.

■ **KEY INSIGHT:** Even functions satisfy $f(-x) = f(x)$. The problem's condition requires $f(3) - f(-3) = 12$, but this difference is always 0.

■ **FAST METHOD:** cx^2 is even $\rightarrow f(3) = f(-3)$ always $\rightarrow f(3) = f(-3) + 12 \rightarrow 0 = 12$. Impossible.

Q16. Correct Answer: B [MEDIUM]

Context: Table: (1,7)(3,13)(5,19)(7,25); find $f(10)$

STEP-BY-STEP SOLUTION:

1. Slope = $(13 - 7) / (3 - 1) = 6 / 2 = 3$.
2. Using point (1, 7): $7 = 3(1) + b \rightarrow b = 4$.
3. $f(x) = 3x + 4$.
4. $f(10) = 3(10) + 4 = 34$.

WHY EACH OPTION IS WRONG:

- A) 31 — Added 6 three times from 25 (as if going $x = 7 \rightarrow 8 \rightarrow 9 \rightarrow 10$ with step +6 each time).
- B) 34 — CORRECT: $f(10) = 3(10) + 4 = 34$. ✓
- C) 37 — Doubled one step (added 12 instead of 9).
- D) 40 — Used slope = 5 instead of 3.

■ **TRAP ALERT:** The table increases by 6 each time x increases by 2 — so the slope is 3 per unit, not 6.

■ **KEY INSIGHT:** Slope = $\Delta y / \Delta x$. Always divide the change in y by the corresponding change in x .

■ **FAST METHOD:** Slope = 3. Shortcut: $f(10) = f(7) + 3(10 - 7) = 25 + 9 = 34$.

Q17. Correct Answer: A [MEDIUM]

Context: $f(x) = 2x + 1$; $f(g(x)) = 2x + 9$; find $g(x)$

STEP-BY-STEP SOLUTION:

1. Write out $f(g(x))$: $f(g(x)) = 2 \cdot g(x) + 1 = 2x + 9$.
2. Isolate $g(x)$: $2 \cdot g(x) = 2x + 8$.
3. $g(x) = x + 4$.

WHY EACH OPTION IS WRONG:

- A) $g(x) = x + 4$ — CORRECT: $2(x+4)+1 = 2x+8+1 = 2x+9$. ✓
- B) $g(x) = x + 8$ — Forgot to account for the +1 in f : gives $2x + 17$.
- C) $g(x) = 2x + 4$ — Gives $f(g(x)) = 4x + 9 \neq 2x + 9$.
- D) $g(x) = 4x + 4$ — Coefficient mismatch on x .

■ **TRAP ALERT:** Dividing by 2 before subtracting the +1 from f . Subtract the constant first: $2g = 2x + 8$, then $g = x + 4$.

■ **KEY INSIGHT:** To find g from $f(g(x)) = \text{expression}$, substitute $g(x)$ as the input of f , set equal, and reverse-engineer.

■ **FAST METHOD:** f adds $+1$. So its input must give $2x + 8$. Thus $g(x) = x + 4$.

Q18. Correct Answer: B [MEDIUM]

Context: $d(r) = \pi r^2$; $d(r+1) - d(r) = 7\pi$; find r

STEP-BY-STEP SOLUTION:

1. Expand $d(r+1)$: $\pi(r+1)^2 = \pi(r^2 + 2r + 1)$.
2. Compute the difference: $\pi(r^2 + 2r + 1) - \pi r^2 = \pi(2r + 1)$.
3. Set equal to 7π : $2r + 1 = 7 \rightarrow 2r = 6 \rightarrow r = 3$.

WHY EACH OPTION IS WRONG:

- A) 2 — Solved $2r = 7$ (ignoring the $+1$): gives $r = 3.5$, incorrect path.
- B) 3 — CORRECT: $\pi(2 \cdot 3 + 1) = 7\pi$. ✓
- C) 4 — Arithmetic error in expanding $(r+1)^2$.
- D) 5 — Further arithmetic error.

■ **TRAP ALERT:** Expanding $(r+1)^2$ as $r^2 + 1$ instead of $r^2 + 2r + 1$ (forgot the middle term).

■ **KEY INSIGHT:** Always expand squared binomials fully: $(r+1)^2 = r^2 + 2r + 1$.

■ **FAST METHOD:** Difference $= \pi(2r + 1) = 7\pi \rightarrow 2r + 1 = 7 \rightarrow r = 3$.

Q19. Correct Answer: B [EASY]

Context: $q(x) = -x + 8$; find $q(-3)$

STEP-BY-STEP SOLUTION:

1. Substitute $x = -3$: $q(-3) = -(-3) + 8$.
2. $-(-3) = +3$.
3. $3 + 8 = 11$.

WHY EACH OPTION IS WRONG:

- A) 5 — Computed $8 - 3 = 5$ (subtracted instead of adding after double negative).
- B) 11 — CORRECT: $-(-3) + 8 = 3 + 8 = 11$. ✓
- C) -5 — Applied an extra negative sign to the result.
- D) -11 — Kept the original negative: $-3 + 8 = 16$ or similar error.

■ **TRAP ALERT:** A negative applied to a negative value: $-(-3) = +3$. Many students keep -3 .

■ **KEY INSIGHT:** Two negatives multiply to a positive: $-(-3) = +3$.

■ **FAST METHOD:** $q(-3) = -(-3) + 8 = 3 + 8 = 11$.

Q20. Correct Answer: B [EASY]

Context: $f(x) = 6x - 7 = 5$; find x

STEP-BY-STEP SOLUTION:

1. Set $6x - 7 = 5$.
2. Add 7 to both sides: $6x = 12$.
3. Divide by 6: $x = 2$.

WHY EACH OPTION IS WRONG:

- A)** 1 — Verify: $6(1) - 7 = -1 \neq 5$.
- B)** 2 — CORRECT: $6(2) - 7 = 12 - 7 = 5$. ✓
- C)** 3 — Verify: $6(3) - 7 = 11 \neq 5$.
- D)** 4 — Divided 12 by 3 instead of 6.

■ **TRAP ALERT:** Dividing by 7 or 3 instead of 6 after isolating $6x = 12$.

■ **KEY INSIGHT:** Inverse operations: add 7 to both sides, then divide by 6.

■ **FAST METHOD:** $6x = 12 \rightarrow x = 2$.

Q21. Correct Answer: A [MEDIUM]

Context: $w(x) = |2x - 6|$; smallest value of $w(x)$?

STEP-BY-STEP SOLUTION:

1. Absolute value is always ≥ 0 .
2. The minimum output is 0, achieved when $2x - 6 = 0 \rightarrow x = 3$.
3. So the smallest value of $w(x)$ is 0.

WHY EACH OPTION IS WRONG:

- A)** 0 — CORRECT: $|2(3) - 6| = |0| = 0$. ✓
- B)** 3 — This is the x -value that produces the minimum, not the minimum output itself.
- C)** 6 — This is the constant inside the expression, not the output.
- D)** -3 — Absolute value can never be negative.

■ **TRAP ALERT:** Confusing the x -value at the minimum ($x = 3$) with the minimum output value (0).

■ **KEY INSIGHT:** Absolute value always outputs non-negative numbers. Its smallest possible value is 0.

■ **FAST METHOD:** Set $2x - 6 = 0 \rightarrow$ minimum output = 0.

Q22. Correct Answer: B [MEDIUM]

Context: $k(x) = 3x + 2$; $k(m) = k(2m - 1)$; find m

STEP-BY-STEP SOLUTION:

1. Set $3m + 2 = 3(2m - 1) + 2$.
2. Expand the right side: $3m + 2 = 6m - 3 + 2 = 6m - 1$.
3. Rearrange: $3 = 3m \rightarrow m = 1$.

WHY EACH OPTION IS WRONG:

- A)** -3 — Sign error when collecting terms.
- B)** 1 — CORRECT: $k(1) = 5$ and $k(2(1)-1) = k(1) = 5$. ✓
- C)** 3 — Divided by the wrong coefficient.
- D)** -1 — Arithmetic error in combining like terms.

■ **TRAP ALERT:** Failing to distribute the 3 across $(2m - 1)$: $3(2m - 1) = 6m - 3$, not $6m - 1$.

■ **KEY INSIGHT:** When $k(m) = k(\text{expression})$ for a one-to-one function, the inputs must be equal: $m = 2m - 1$.

■ **FAST METHOD:** Inputs must match: $m = 2m - 1 \rightarrow 1 = m$.

Q23. Correct Answer: D [HARD]

Context: $f(x)=3x-1$; $g(x)=x+5$; solve $f(g(x))=g(f(x))$

STEP-BY-STEP SOLUTION:

1. Compute $f(g(x))$: $f(x+5) = 3(x+5) - 1 = 3x + 15 - 1 = 3x + 14$.
2. Compute $g(f(x))$: $g(3x-1) = (3x-1) + 5 = 3x + 4$.
3. Set equal: $3x + 14 = 3x + 4$.
4. Subtract $3x$ from both sides: $14 = 4$. This is a contradiction.
5. No value of x satisfies the equation.

WHY EACH OPTION IS WRONG:

- A)** All real values — Would require $14 = 4$ to be true for all x . It is not.
- B)** $x = 1$ — Check: $f(g(1)) = 3(6)-1 = 17$; $g(f(1)) = g(2) = 7$. $17 \neq 7$.
- C)** $x = 6$ — Check: $f(g(6)) = 3(11)-1 = 32$; $g(f(6)) = g(17) = 22$. $32 \neq 22$.
- D)** No solution exists — CORRECT: $3x + 14 = 3x + 4$ gives $14 = 4$, impossible. ✓

■ **TRAP ALERT:** When the x -terms cancel and leave a false statement ($14 = 4$), there is NO solution — not infinitely many.

■ **KEY INSIGHT:** If simplification gives False statement $\rightarrow$ no solution. If True statement ($0 = 0$) $\rightarrow$ all x . If equation in $x \rightarrow$ one solution.

■ **FAST METHOD:** Compute both. x -terms cancel. Constants: $14 \neq 4$. No solution.

Q24. Correct Answer: A [HARD]

Context: $f(x)=x^2+bx+c$; $f(2)=0$; $f(5)=0$; find $f(0)$

STEP-BY-STEP SOLUTION:

1. $f(2) = 0$ and $f(5) = 0$ mean $x = 2$ and $x = 5$ are roots.
2. So $f(x) = (x - 2)(x - 5) = x^2 - 7x + 10$.
3. $f(0) = (0 - 2)(0 - 5) = (-2)(-5) = 10$.

WHY EACH OPTION IS WRONG:

- A)** 10 — CORRECT: $(-2)(-5) = 10$. ✓
- B)** -10 — Sign error: $(-2)(-5) = +10$, not -10.
- C)** 7 — This is $|b|$, the absolute value of the x-coefficient.
- D)** -7 — This is $b = -7$ (the x-coefficient), not the value of $f(0)$.

■ **TRAP ALERT:** Getting the sign wrong on the product $(-2)(-5)$. Two negatives multiply to a positive.

■ **KEY INSIGHT:** If roots are r and s , $f(x) = (x-r)(x-s)$. Then $f(0) = (0-r)(0-s) = r \times s =$ product of the roots.

■ **FAST METHOD:** $f(0) = (-2)(-5) = 10$. Lightning-fast using the factored form.

Q25. Correct Answer: C [HARD]

Context: $f(x) = (2x+1)/(x-3)$; find $f(f(5))$

STEP-BY-STEP SOLUTION:

1. Step 1: $f(5) = (2 \cdot 5 + 1)/(5 - 3) = 11/2$.
2. Step 2: $f(11/2) = (2 \cdot (11/2) + 1)/((11/2) - 3) = (11 + 1)/(11/2 - 6/2) = 12/(5/2) = 12 \times 2/5 = 24/5$.

WHY EACH OPTION IS WRONG:

- A)** 5 — This is the input, not the output of $f(f(5))$.
- B)** $11/2$ — This is $f(5)$, the result after only one application of f .
- C)** $24/5$ — CORRECT: $f(5) = 11/2$, then $f(11/2) = 24/5$. ✓
- D)** Undefined — $f(5) = 11/2$ (denominator $2 \neq 0$); $f(11/2)$: denominator $= 5/2 \neq 0$. Defined.

■ **TRAP ALERT:** Stopping after the first application of f (getting $11/2 = f(5)$) instead of applying f a second time.

■ **KEY INSIGHT:** $f(f(5))$ means apply f TWICE. Compute the inner $f(5)$ first, then plug that result back into f .

■ **FAST METHOD:** $f(5) = 11/2$. Then $f(11/2) = 12 \div (5/2) = 24/5$.

SECTION B — CREATING LINEAR MODELS FROM WORD PROBLEMS (Q26–Q50)

Q26. Correct Answer: C [EASY]

Context: Taxi: \$3.00 flat + \$2.50/mile

STEP-BY-STEP SOLUTION:

1. Fixed flat fee = \$3.00 → y-intercept (does not change with miles).
2. Per-mile rate = \$2.50 → slope.
3. $C = 2.50m + 3.00$.

WHY EACH OPTION IS WRONG:

- A)** $C = 2.50m$ — Missing the flat fee entirely.
- B)** $C = 3.00m + 2.50$ — Swapped slope and intercept.
- C)** $C = 2.50m + 3.00$ — CORRECT: rate $\times$ miles + flat fee. ✓
- D)** $C = 3.00 + 2.50 + m$ — Treats m as addition, not multiplication by a rate.

■ **TRAP ALERT:** Swapping which value is the rate (slope) and which is the flat fee (intercept).

■ **KEY INSIGHT:** What changes per unit → slope. What stays fixed regardless of quantity → y-intercept.

■ **FAST METHOD:** $C = 2.50m + 3.00$.

Q27. Correct Answer: C [EASY]

Context: Candle: 18 cm tall, loses 2.4 cm/hour

STEP-BY-STEP SOLUTION:

1. Starting height = 18 cm → y-intercept.
2. 'Loses' means decreasing → subtract.
3. $h = 18 - 2.4t$.

WHY EACH OPTION IS WRONG:

- A)** $h = 2.4t + 18$ — Adds instead of subtracts; height would increase.
- B)** $h = 18t - 2.4$ — Treats starting height as the rate, which is wrong.
- C)** $h = 18 - 2.4t$ — CORRECT: starts at 18, decreases 2.4 per hour. ✓
- D)** $h = 2.4 - 18t$ — Tiny intercept and enormous negative rate; nonsensical.

■ **TRAP ALERT:** When something is losing, decreasing, or burning, the rate must be subtracted.

■ **KEY INSIGHT:** DSAT word signals: 'loses,' 'decreases,' 'drains,' 'falls' → negative slope. 'Gains,' 'earns,' 'grows' → positive slope.

■ **FAST METHOD:** Start – rate $\times$ time = $18 - 2.4t$.

Q28. Correct Answer: D [EASY]

Context: Marcus: \$12.50/hr + \$75 one-time bonus

STEP-BY-STEP SOLUTION:

1. Per-hour rate = \$12.50 → slope.
2. One-time bonus = \$75 → y-intercept (paid once, not per hour).

3. $E = 12.50h + 75$.

WHY EACH OPTION IS WRONG:

- A) $E = 12.50h$ — Missing the signing bonus.
- B) $E = 75h + 12.50$ — Swapped: \$75 is not a per-hour rate.
- C) $E = 12.50 + 75h$ — \$75 is not a per-hour charge.
- D) $E = 12.50h + 75$ — CORRECT: hourly earnings + one-time bonus. ✓

■ **TRAP ALERT:** Confusing a one-time fixed amount (intercept) with a per-unit rate (slope).

■ **KEY INSIGHT:** 'One-time' or 'flat' → y-intercept. 'Per [unit]' → slope.

■ **FAST METHOD:** $E = 12.50h + 75$.

Q29. Correct Answer: B [EASY]

Context: Tank: 240 gal, drains 15 gal/min

STEP-BY-STEP SOLUTION:

1. Starting volume = 240 gallons → y-intercept.
2. 'Drains' means decreasing → subtract.
3. $V = 240 - 15m$.

WHY EACH OPTION IS WRONG:

- A) $V = 15m$ — Represents volume drained, not volume remaining.
- B) $V = 240 - 15m$ — CORRECT. ✓
- C) $V = 15 - 240m$ — Absurdly small starting value and huge negative rate.
- D) $V = 240m - 15$ — Rate and intercept reversed.

■ **TRAP ALERT:** Writing volume drained (15m) instead of volume remaining (240 – 15m).

■ **KEY INSIGHT:** Remaining = Start – (rate × time). One of the most frequent DSAT linear model patterns.

■ **FAST METHOD:** $V = 240 - 15m$.

Q30. Correct Answer: C [EASY]

Context: Gym: \$40 enrollment + \$25/month

STEP-BY-STEP SOLUTION:

1. Monthly fee = \$25 → slope.
2. Enrollment fee = \$40 (paid once) → y-intercept.
3. $C = 25m + 40$.

WHY EACH OPTION IS WRONG:

- A) $C = 65m$ — Combined both fees as one per-month charge.

- B)** $C = 40m + 25$ — Enrollment fee is one-time, not \$40/month.
- C)** $C = 25m + 40$ — CORRECT. ✓
- D)** $C = 40 + 25 + m$ — Treats m as a fee rather than the count of months.

■ **TRAP ALERT:** Adding both fees as if both are monthly costs: $40 + 25 = \$65$ per month.

■ **KEY INSIGHT:** One-time/flat fees = intercept. Recurring per-unit charges = slope.

■ **FAST METHOD:** $C = 25m + 40$.

Q31. Correct Answer: **B** [MEDIUM]

Context: Plumber: \$85 first hr, \$60 each additional hr ($h \geq 1$)

STEP-BY-STEP SOLUTION:

1. First hour: \$85 (fixed for that unit).
2. Additional hours: $(h - 1)$ hours $\times$ \$60.
3. $T = 85 + 60(h - 1) = 85 + 60h - 60 = 60h + 25$.
4. Verify at $h=1$: $60(1)+25 = 85$. ✓ At $h=2$: $60(2)+25 = 145 = 85+60$. ✓

WHY EACH OPTION IS WRONG:

- A)** $T = 60h + 85$ — Charges \$60 for every hour including the first.
- B)** $T = 60h + 25$ — CORRECT: $60(h-1)+85$ simplifies to $60h+25$. ✓
- C)** $T = 85h + 60$ — Treats \$85 as the per-hour rate.
- D)** $T = 145h$ — Simply adds both rates together as a single rate.

■ **TRAP ALERT:** Writing $60h + 85$ charges \$60 for the first hour too, but the first hour costs \$85.

■ **KEY INSIGHT:** When the rate changes after the first unit: $T = [\text{first unit cost}] + [\text{rate} \times (h - 1)]$. Simplify to find the true intercept.

■ **FAST METHOD:** Plug in $h=1$: T must equal 85. Check B: $60(1)+25=85$. ✓ Check A: $60+85=145$. ✗

Q32. Correct Answer: **A** [MEDIUM]

Context: Freezer: 8°F , cools 3°F per 10 min; when does it reach -4°F ?

STEP-BY-STEP SOLUTION:

1. Convert rate: 3°F per 10 min = 0.3°F per minute.
2. $T(x) = 8 - 0.3x$.
3. Set equal to -4 : $8 - 0.3x = -4 \rightarrow 0.3x = 12 \rightarrow x = 40$ minutes.

WHY EACH OPTION IS WRONG:

- A)** $T = 8 - 0.3x$; 40 minutes — CORRECT. ✓
- B)** $T = 8 + 0.3x$; 40 minutes — Wrong sign: cooling means temperature decreases.
- C)** $T = 8 - 3x$; 4 minutes — Used $3^\circ\text{F}/\text{min}$ instead of 3°F per 10 min.

D) $T = 8 - 0.3x$; 30 minutes — Equation correct but arithmetic wrong: $0.3(30) = 9$, giving $T = -1 \neq -4$.

■ **TRAP ALERT:** Using 3 as the per-minute rate instead of converting: 3°F per 10 minutes $= 0.3^\circ\text{F}$ per minute.

■ **KEY INSIGHT:** Always convert the rate to match the units of your variable before writing the equation.

■ **FAST METHOD:** Rate $= 3/10 = 0.3/\text{min}$. Then $8 - 0.3x = -4 \rightarrow x = 40$.

Q33. Correct Answer: **A** [MEDIUM]

Context: Alicia: $6+3h$ miles; Ben: $4.5h$ miles

STEP-BY-STEP SOLUTION:

1. Alicia's total after h hours: $6 + 3h$.
2. Ben's total after h hours: $4.5h$.
3. Set equal: $6 + 3h = 4.5h$.

WHY EACH OPTION IS WRONG:

- A)** $3h + 6 = 4.5h$ — CORRECT. ✓
- B)** $3h = 4.5h + 6$ — Incorrectly gives Ben the 6-mile head start.
- C)** $6h + 3 = 4.5h$ — Swapped the role of 6 and 3.
- D)** $4.5h + 6 = 3h$ — Gives Ben the head start AND wrong direction.

■ **TRAP ALERT:** Assigning the 6-mile head start to the wrong person. Alicia has the head start, not Ben.

■ **KEY INSIGHT:** The person who already has progress gets the added constant. The person starting from zero has no constant term.

■ **FAST METHOD:** Alicia's distance = Ben's distance: $6 + 3h = 4.5h$.

Q34. Correct Answer: **B** [MEDIUM]

Context: Phone: $\$0.05/\text{text} + \29.99 monthly; total= $\$34.99$

STEP-BY-STEP SOLUTION:

1. Total cost = variable cost + fixed cost.
2. $0.05t + 29.99 = 34.99$.

WHY EACH OPTION IS WRONG:

- A)** $0.05t = 34.99$ — Ignores the $\$29.99$ monthly fee.
- B)** $0.05t + 29.99 = 34.99$ — CORRECT: variable + fixed = total. ✓
- C)** $0.05 + 29.99t = 34.99$ — Swapped per-text and monthly charges.
- D)** $29.99t - 0.05 = 34.99$ — Nonsensical structure.

■ **TRAP ALERT:** Forgetting to include the flat monthly fee in the total cost equation.

■ **KEY INSIGHT:** Total cost = variable cost + fixed cost. Always include both components.

■ **FAST METHOD:** $0.05t + 29.99 = 34.99 \rightarrow 0.05t = 5 \rightarrow t = 100$ texts.

Q35. Correct Answer: B [MEDIUM]

Context: Printing: \$0.08/page for first 500; \$0.05/page after

STEP-BY-STEP SOLUTION:

1. Cost for first 500 pages: $500 \times \$0.08 = \40 (fixed).
2. Additional pages beyond 500: $(n - 500)$ pages $\times \$0.05$.
3. Total: $C = 40 + 0.05(n - 500) = 40 + 0.05n - 25 = 0.05n + 15$.

WHY EACH OPTION IS WRONG:

- A) $C = 0.08n$ — Charges \$0.08 for all n pages, ignoring the reduced rate after 500.
- B) $C = 0.05n + 15$ — CORRECT: simplification of $40 + 0.05(n-500)$. ✓
- C) $C = 0.08n + 0.05(n-500)$ — Double-counts the first 500 pages at both rates.
- D) $C = 0.08(n-500) + 40$ — Applies the first-tier rate to excess pages (wrong rate for excess).

■ **TRAP ALERT:** Option C double-counts: it charges \$0.08 for all n pages AND adds \$0.05 for pages over 500.

■ **KEY INSIGHT:** For two-rate models: compute the fixed cost for the threshold quantity, then add rate $\times$ excess.

■ **FAST METHOD:** First 500 pages = \$40. Excess = $0.05(n-500)$. Simplify: $0.05n + 15$.

Q36. Correct Answer: C [MEDIUM]

Context: $P=3.2t+5.6$; what does 5.6 represent?

STEP-BY-STEP SOLUTION:

1. In $P = mt + b$, the constant b is the y -intercept.
2. At $t = 0$: $P = 3.2(0) + 5.6 = 5.6$.
3. 5.6 = initial population (in thousands) when the experiment began.

WHY EACH OPTION IS WRONG:

- A) Rate of increase — That is 3.2 (the slope), not 5.6.
- B) Population after one hour — At $t=1$: $P = 3.2+5.6 = 8.8$, not 5.6.
- C) Initial bacteria population, in thousands — CORRECT: value at $t = 0$. ✓
- D) Hours to reach 5,600 — 5,600 thousands is not directly found this way.

■ **TRAP ALERT:** Confusing the y -intercept (initial value) with the slope (rate of change).

■ **KEY INSIGHT:** In $P = mt + b$: b is the INITIAL VALUE (when $t = 0$). m is the RATE OF CHANGE per unit time.

■ **FAST METHOD:** Set $t = 0$: $P = 5.6$. The constant is always the starting value.

Q37. Correct Answer: C [MEDIUM]

Context: $C = 0.22d + 35$; interpret 0.22

STEP-BY-STEP SOLUTION:

1. 0.22 is the coefficient of d (miles driven).
2. Slope = rate of change of C per mile driven.
3. 0.22 = cost per mile, in dollars.

WHY EACH OPTION IS WRONG:

- A) Fixed daily rental fee — That is \$35 (the y -intercept).
- B) Total cost after 1 mile — That is $C(1) = 0.22 + 35 = \$35.22$.
- C) Cost per mile driven, in dollars — CORRECT: slope represents rate per mile. ✓
- D) Miles included before additional charges — No such threshold in this model.

■ **TRAP ALERT:** Confusing the slope (0.22) with the intercept (35). The slope is always the per-unit cost.

■ **KEY INSIGHT:** In $y = mx + b$: m = rate per unit (slope); b = fixed starting amount (intercept).

■ **FAST METHOD:** Slope = how much C changes per additional mile = \$0.22.

Q38. Correct Answer: B [MEDIUM]

Context: Electrician: \$95/hr; \$230 materials; net $\geq \$1,000$

STEP-BY-STEP SOLUTION:

1. Gross earnings from working h hours: $95h$.
2. Net earnings after deducting materials: $95h - 230$.
3. Condition: $95h - 230 \geq 1,000$.

WHY EACH OPTION IS WRONG:

- A) $95h \geq 1,000$ — Ignores the \$230 materials deduction.
- B) $95h - 230 \geq 1,000$ — CORRECT: net = hourly earnings minus material cost. ✓
- C) $95h + 230 \geq 1,000$ — Adds the cost instead of subtracting it.
- D) $95 - 230h \geq 1,000$ — Treats \$230 as a per-hour charge, which it is not.

■ **TRAP ALERT:** Adding materials cost instead of subtracting. She pays for materials, so it reduces her net earnings.

■ **KEY INSIGHT:** Net = Revenue – Expenses. Deductions are always subtracted.

■ **FAST METHOD:** $\text{Net} = 95h - 230 \geq 1,000$.

Q39. Correct Answer: C [HARD]

Context: Company A: \$120/mo+\$50. Company B: \$90/mo+\$170. Equal when?

STEP-BY-STEP SOLUTION:

1. Company A total: $120m + 50$.
2. Company B total: $90m + 170$.
3. Set equal: $120m + 50 = 90m + 170$.
4. $30m = 120 \rightarrow m = 4$ months.
5. Verify: $A = 120(4) + 50 = 530$; $B = 90(4) + 170 = 530$. ✓

WHY EACH OPTION IS WRONG:

- A) 2 — $A = 290$, $B = 350$. Not equal.
- B) 3 — $A = 410$, $B = 440$. Not equal.
- C) 4 — CORRECT: $A = B = 530$. ✓
- D) 6 — $A = 770$, $B = 710$. Company A is more expensive.

■ **TRAP ALERT:** Subtracting the registration fees from the wrong side, or using wrong signs.

■ **KEY INSIGHT:** Registration fees are added (not subtracted) to the monthly costs when setting up the total cost expressions.

■ **FAST METHOD:** $120m + 50 = 90m + 170 \rightarrow 30m = 120 \rightarrow m = 4$.

Q40. Correct Answer: A [HARD]

Context: Widgets: 840 at 9AM, 1160 at 1PM; find $W(t)$

STEP-BY-STEP SOLUTION:

1. Elapsed time: $1\text{PM} - 9\text{AM} = 4$ hours.
2. Change in widgets: $1,160 - 840 = 320$.
3. Rate = $320 / 4 = 80$ widgets per hour.
4. At $t = 0$ (9AM): $W = 840$.
5. $W(t) = 80t + 840$.

WHY EACH OPTION IS WRONG:

- A) $W(t) = 80t + 840$ — CORRECT. ✓
- B) $W(t) = 80t + 760$ — Wrong intercept: shifted one hour back.
- C) $W(t) = 320t + 840$ — Used total change (320) as the per-hour rate.
- D) $W(t) = 80t + 1,160$ — Used the 1PM count as the y-intercept.

■ **TRAP ALERT:** Using the total change (320) as the rate instead of dividing by time elapsed (4 hours) to get 80 per hour.

■ **KEY INSIGHT:** Rate = total change ÷ time elapsed. Then use any given data point to find the intercept.

■ **FAST METHOD:** Rate = $320/4 = 80$. At $t=0$, $W=840$. So $W(t) = 80t + 840$.

Q41. Correct Answer: A [HARD]

Context: Equipment: \$8,400 at purchase; \$5,100 after 6 years. Interpret slope.

STEP-BY-STEP SOLUTION:

1. Change in value: $5,100 - 8,400 = -3,300$.
2. Time elapsed: 6 years.
3. Slope = $-3,300 / 6 = -550$ per year.
4. The equipment loses \$550 in value each year.

WHY EACH OPTION IS WRONG:

- A) Loses \$550 per year — CORRECT: slope = -550 . ✓
- B) Loses \$5,100 per year — \$5,100 is the value after 6 years, not the annual depreciation.
- C) Purchased for \$8,400 — This is the y-intercept (initial value), not the slope.
- D) Worth \$0 after ~15.3 years — This is the x-intercept, not the slope interpretation.

■ **TRAP ALERT:** Confusing the slope (rate of depreciation per year) with the y-intercept (purchase price) or x-intercept.

■ **KEY INSIGHT:** Slope in a depreciation model = annual change in value. Negative slope means 'loses value each year.'

■ **FAST METHOD:** Slope = $(5,100 - 8,400)/6 = -550/\text{year} \rightarrow$ loses \$550 per year.

Q42. Correct Answer: C [HARD]

Context: Cafeteria: \$3.75/lunch; \$210 daily costs; minimum lunches?

STEP-BY-STEP SOLUTION:

1. Revenue must cover costs: $3.75n \geq 210$.
2. $n \geq 210 / 3.75 = 56$.
3. The minimum is 56 lunches.
4. Verify: $3.75 \times 56 = 210.00$ (costs exactly covered). ✓

WHY EACH OPTION IS WRONG:

- A) 54 — $3.75 \times 54 = \$202.50 < \210 . Not enough.
- B) 55 — $3.75 \times 55 = \$206.25 < \210 . Not enough.
- C) 56 — CORRECT: $3.75 \times 56 = \$210.00$. Exactly covers costs. ✓
- D) 57 — This covers costs but is not the minimum.

■ **TRAP ALERT:** Rounding 56 down to 55 because $210 \div 3.75 = 56$ exactly and students fear whole number answers.

■ **KEY INSIGHT:** Break-even: revenue = cost. Check that the computed value exactly satisfies the condition.

■ **FAST METHOD:** $210 \div 3.75 = 56$ exactly. Minimum = 56.

Q43. Correct Answer: **C** [HARD]

Context: Whales: 1,240 in 2010, decrease 48/year; first year below 800?

STEP-BY-STEP SOLUTION:

1. Model: $P = 1,240 - 48y$, where y = years after 2010.
2. Set $P < 800$: $1,240 - 48y < 800$.
3. $48y > 440 \rightarrow y > 9.167$.
4. First whole year when $y > 9.167$ is $y = 10$.
5. Year = $2010 + 10 = 2020$.
6. Verify: $P(10) = 1,240 - 480 = 760 < 800$. ✓ $P(9) = 1,240 - 432 = 808 > 800$. ✗

WHY EACH OPTION IS WRONG:

- A)** 2018 — $y = 8$: $P = 856 > 800$. Not below yet.
- B)** 2019 — $y = 9$: $P = 808 > 800$. Still not below 800.
- C)** 2020 — CORRECT: $y = 10$: $P = 760 < 800$. First year below. ✓
- D)** 2021 — Population is already below 800 in 2020.

■ **TRAP ALERT:** Using $y > 9.167$ and rounding down to $y = 9$ (year 2019) instead of up to $y = 10$ (year 2020).

■ **KEY INSIGHT:** Strict inequality with 'first year': round UP to the next whole year when the decimal doesn't resolve cleanly.

■ **FAST METHOD:** $y > 440/48 = 9.17 \rightarrow y = 10 \rightarrow 2010 + 10 = 2020$.

Q44. Correct Answer: **C** [HARD]

Context: Movers: $b + r/\text{mile}$; 20 mi=\$150; 35 mi=\$195. Find b .

STEP-BY-STEP SOLUTION:

1. Equation 1: $b + 20r = 150$.
2. Equation 2: $b + 35r = 195$.
3. Subtract Eq1 from Eq2: $15r = 45 \rightarrow r = 3$.
4. Substitute into Eq1: $b + 60 = 150 \rightarrow b = 90$.
5. Verify: $90 + 35(3) = 90 + 105 = 195$. ✓

WHY EACH OPTION IS WRONG:

- A) \$30 — Used wrong rate ($r = 6$ instead of $r = 3$).
- B) \$60 — This is $20r = 20 \times 3$ (the per-mile contribution), not the base fee.
- C) \$90 — CORRECT: $b = 150 - 60 = 90$. ✓
- D) \$120 — Arithmetic error in the subtraction step.

■ **TRAP ALERT:** Subtracting equations in the wrong direction, or confusing the per-mile contribution ($20r = 60$) for the base fee.

■ **KEY INSIGHT:** Eliminate the base fee by subtracting the two equations to isolate r first, then solve for b .

■ **FAST METHOD:** Difference: $195 - 150 = 45$ for 15 extra miles $\rightarrow r = 3$. Then $b = 150 - 20(3) = 90$.

Q45. Correct Answer: B [HARD]

Context: Tank A: $120 + 8t$; Tank B: $105 + 11t$; when $B = A + 15$?

STEP-BY-STEP SOLUTION:

1. Tank A after t minutes: $120 + 8t$.
2. Tank B after t minutes: $105 + 11t$.
3. Set $B = A + 15$: $105 + 11t = (120 + 8t) + 15$.
4. $105 + 11t = 135 + 8t$.
5. $3t = 30 \rightarrow t = 10$ minutes.
6. Verify: $B = 105 + 11(10) = 215$; $A = 120 + 8(10) = 200$; $B - A = 15$. ✓

WHY EACH OPTION IS WRONG:

- A) 5 min — $B = 160$, $A = 160$. $B - A = 0$, not 15.
- B) 10 min — CORRECT: $B = 215$, $A = 200$, $B - A = 15$. ✓
- C) 15 min — $B = 270$, $A = 240$, $B - A = 30 \neq 15$.
- D) 20 min — $B = 325$, $A = 280$, $B - A = 45 \neq 15$.

■ **TRAP ALERT:** Setting up the inequality in the wrong direction. 'B contains 15 more than A' means $B = A + 15$, not $A = B + 15$.

■ **KEY INSIGHT:** Read the problem carefully: 'Tank B contains exactly 15 liters more than Tank A' means $B - A = 15$, equivalently $B = A + 15$.

■ **FAST METHOD:** $105 + 11t = 135 + 8t \rightarrow 3t = 30 \rightarrow t = 10$.

Q46. Correct Answer: C [MEDIUM]

Context: $S = -3w + 74$; what does 74 represent?

STEP-BY-STEP SOLUTION:

1. In $S = -3w + 74$, the constant 74 is the y-intercept.
2. At $w = 0$: $S = 74$.

3. 74 = number of students enrolled when the program began (at week 0).

WHY EACH OPTION IS WRONG:

- A) Students leaving each week — That is 3 (the magnitude of the slope, -3).
- B) Students after 74 weeks — This misinterprets 74 as a time value.
- C) Students enrolled when program began — CORRECT: $w = 0$ gives $S = 74$. ✓
- D) Week when enrollment reaches zero — That is when $S = 0$: $w \approx 24.7$, not 74.

■ **TRAP ALERT:** Confusing the y-intercept (74, the starting enrollment) with the x-intercept (when enrollment = 0).

■ **KEY INSIGHT:** The constant in a linear model = initial value when the variable equals zero.

■ **FAST METHOD:** Set $w = 0$: $S = 74$. The constant is always the starting value.

Q47. Correct Answer: B [MEDIUM]

Context: Driver: \$0.40/mile + \$80 daily; paid \$180; find miles

STEP-BY-STEP SOLUTION:

1. Set up: $0.40m + 80 = 180$.
2. Subtract 80: $0.40m = 100$.
3. Divide by 0.40: $m = 250$ miles.
4. Verify: $0.40(250) + 80 = 100 + 80 = 180$. ✓

WHY EACH OPTION IS WRONG:

- A) 200 miles — $0.40(200) + 80 = 80 + 80 = 160 \neq 180$.
- B) 250 miles — CORRECT: $0.40(250) + 80 = 180$. ✓
- C) 300 miles — $0.40(300) + 80 = 120 + 80 = 200 \neq 180$.
- D) 450 miles — Divided 180 by 0.40 without first subtracting \$80.

■ **TRAP ALERT:** Dividing the total (\$180) by the per-mile rate (\$0.40) without first subtracting the \$80 daily guarantee.

■ **KEY INSIGHT:** Isolate the variable term first: subtract the fixed amount, then divide by the rate.

■ **FAST METHOD:** $0.40m = 180 - 80 = 100 \rightarrow m = 250$.

Q48. Correct Answer: A [MEDIUM]

Context: Pool: 4,400 gal at $t=2$; 2,900 gal at $t=5$

STEP-BY-STEP SOLUTION:

1. Slope = $(2,900 - 4,400) / (5 - 2) = -1,500 / 3 = -500$ gal/hr.
2. Use point (2, 4,400): $4,400 = -500(2) + b \rightarrow b = 5,400$.
3. $G = -500t + 5,400$.
4. Verify: $G(2) = -1,000 + 5,400 = 4,400$. ✓ $G(5) = -2,500 + 5,400 = 2,900$. ✓

WHY EACH OPTION IS WRONG:

- A) $G = -500t + 5,400$ — CORRECT. ✓
- B) $G = -500t + 4,400$ — Used the point (2, 4,400) as if $t = 0$; wrong intercept.
- C) $G = 500t + 5,400$ — Positive slope means filling; pool is draining.
- D) $G = -1,500t + 7,400$ — Used total change (-1,500) as the rate, not divided by 3.

■ **TRAP ALERT:** Using the given data point (2, 4,400) directly as the y-intercept without finding the true y-intercept at $t = 0$.

■ **KEY INSIGHT:** Find slope first using two points, then substitute one point into $G = mt + b$ to solve for b .

■ **FAST METHOD:** Slope = -500. Sub (2, 4400): $b = 4400 + 1000 = 5400$. $G = -500t + 5400$.

Q49. Correct Answer: B [HARD]

Context: 10 units → \$860; 25 units → \$1,160; find earnings at 40 units

STEP-BY-STEP SOLUTION:

1. Commission per unit = $(1,160 - 860) / (25 - 10) = 300 / 15 = \20 .
2. Base salary: $860 = 20(10) + b \rightarrow b = 860 - 200 = 660$.
3. Earnings model: $E = 20n + 660$.
4. At $n = 40$: $E = 20(40) + 660 = 800 + 660 = \$1,460$.
5. Verify: $E(10) = 200 + 660 = 860$. ✓ $E(25) = 500 + 660 = 1,160$. ✓

WHY EACH OPTION IS WRONG:

- A) \$1,360 — Used $n = 35$ (off by 5 units).
- B) \$1,460 — CORRECT: $20(40) + 660 = 1,460$. ✓
- C) \$1,560 — Used commission rate of \$30 instead of \$20.
- D) \$1,760 — Used wrong base salary.

■ **TRAP ALERT:** Computing the commission rate as $300 \div 10 = \$30$ instead of $300 \div 15 = \$20$. Divide by the CHANGE in units ($25 - 10 = 15$), not by one of the unit values.

■ **KEY INSIGHT:** Rate = change in earnings $\div$ change in units sold. Always use the difference between two data points.

■ **FAST METHOD:** Rate = $300/15 = 20$. Base = 660. At 40 units: $20(40) + 660 = 1,460$.

Q50. Correct Answer: B [HARD]

Context: $P = 1.6t + 42$; pounds in week 8 vs week 3

STEP-BY-STEP SOLUTION:

1. $P(8) = 1.6(8) + 42 = 12.8 + 42 = 54.8$.
2. $P(3) = 1.6(3) + 42 = 4.8 + 42 = 46.8$.

3. Difference = $54.8 - 46.8 = 8$ pounds.
4. Shortcut: difference = slope $\times$ interval = $1.6 \times (8 - 3) = 1.6 \times 5 = 8$.

WHY EACH OPTION IS WRONG:

- A)** 4 pounds — Used wrong interval or halved the slope.
- B)** 8 pounds — CORRECT: $1.6 \times 5 = 8$. ✓
- C)** 16 pounds — Multiplied 1.6×10 (wrong interval).
- D)** 5 pounds — This is the time difference (weeks), not the pound difference.

■ **TRAP ALERT:** For a linear model, the change over an interval depends only on slope $\times$ interval. The constant (+42) cancels out.

■ **KEY INSIGHT:** KEY SHORTCUT: In $P = mt + b$, change from t_1 to $t_2 = m(t_2 - t_1)$. The constant always cancels.

■ **FAST METHOD:** Change = $1.6 \times (8 - 3) = 1.6 \times 5 = 8$. No need to compute individual values.

TOPIC SUMMARIES

TOPIC SUMMARY — Function Notation & Evaluation

Common Mistakes Students Make:

1. Substituting only part of the expression — e.g., computing 7×5 but forgetting $+3$.
2. Reversing composition order: $g(f(x))$ vs $f(g(x))$. Always work INSIDE-OUT.
3. Confusing $f(2x)$ with $2f(x)$ — these are fundamentally different operations.

Must Remember — Top Rules:

1. $f(a)$ = replace every x in the formula with a , then simplify.
2. Composition $f(g(x))$: evaluate the INNER function first, plug into OUTER.
3. Linear functions: $f(x+c) - f(x) = \text{slope} \times c$ (a constant — plug-in is unnecessary).
4. $f(0)$ = the constant term for any polynomial (the y-intercept).
5. Even functions: $f(-x) = f(x)$. Odd: $f(-x) = -f(x)$.

Expected on DSAT:

Typically 2–3 questions per Math section. Appear as Easy/Medium in Module 1 and Hard MCQs in Module 2. Composition and functional equations are the most common Hard sub-types.

TOPIC SUMMARY — Creating Linear Models from Word Problems

Common Mistakes Students Make:

1. Swapping slope and y-intercept — the RATE is the slope; the FIXED amount is the intercept.

2. Not converting rates to matching units (e.g., '3°F per 10 min' → 0.3°F per min).
3. Wrong sign on change — 'decreases,' 'drains,' 'costs' require subtraction from the starting value.

Must Remember — Top Rules:

1. Slope = rate of change per unit. Y-intercept = initial/fixed value (when variable = 0).
2. Remaining = Starting amount – (rate × time).
3. Two-rate models: compute fixed cost for threshold, add rate × excess.
4. Slope interpretation: the slope = 'amount of change per 1 unit increase in x.'
5. Two-point slope: $m = (y_2 - y_1)/(x_2 - x_1)$. Always divide CHANGE in y by CHANGE in x.

Expected on DSAT:

Typically 3–5 questions per Math section — one of the HIGHEST frequency topics. Appears at all difficulty levels. Slope/intercept interpretation (Medium) and two-point model construction (Hard) are especially common.

— END OF SOLUTIONS —

All 50 questions are original · Prepared in authentic Digital SAT style · 11 errors identified and corrected in this final edition